Saman Abeysinghe

AI Engineer

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SUMMARY

AI Engineer with a BSc (Hons) in Engineering, specialised in Computer Engineering. I design and build production-ready AI applications using LLMs, RAG, multi-agent architectures, document intelligence and computer vision. Currently at Cenango (Pvt) Ltd, with earlier experience in backend development, machine learning and embedded systems.

EXPERIENCE

AI Engineer — 2025 – Present
Cenango (Pvt) Ltd
- Developing AI-powered systems built on LLMs, RAG, and multi-agent architectures.

Temporary Demonstrator — 2025
The Open University of Sri Lanka
- Conducted laboratory sessions for Microprocessors & Interfacing, Object Oriented Programming, Software Development for Engineers, and Computer Technology.

Visiting Technician — 2024
The Open University of Sri Lanka
- Conducted laboratory sessions for the Microprocessors & Interfacing course.
- Worked with micro-controller programming and simulation software.

Software Engineer Trainee — 2023 – 2024
The Open University of Sri Lanka
- Implemented backend services using Spring Boot.
- Managed and maintained a PostgreSQL database.
- Wrote C++ code to program micro-controllers.

Research Engineer Trainee — 2022
NERDC
- Implemented C++ programs for micro-controller based projects.
- Used simulation software to run and validate simulations.

SELECTED AI PROJECTS

RAG-Based Chatbot for WordPress Sites
Claude · Voyage AI · Qdrant · Flask · AWS Bedrock · SQS/SNS · PostgreSQL
- Built end to end: a self-serve multi-tenant platform where customers register on a dashboard and connect their WordPress site.
- The system crawls and indexes the site content, then issues a keyed script tag that embeds a Claude-powered chat widget answering visitor questions from that content.
- Content ingestion runs asynchronously through AWS SQS/SNS.

Document & Handwriting Extraction System
YOLO · Claude (AWS Bedrock) · Next.js · Flask · PostgreSQL
- Reads unstructured documents containing handwritten English text, detecting question and answer fields separately and returning structured question-and-answer data.
- Users can edit, search, delete and add extracted fields, and view the fully annotated document.
- A rule engine evaluates each document against user-defined rules and reports which passed and failed.

Fully Local, Private RAG System
Ollama · ChromaDB · React · Flask · PostgreSQL
- Chat system that runs entirely on the user's machine with no internet access, for sectors such as banking and legal work that need a private environment.
- Users upload any documents and query across all of them, manage the document inventory, and rate each response so the system keeps improving.

MACHINE LEARNING PROJECTS

Nutrient Deficiency Detection for Black Pepper Leaf

TensorFlow · Python · Flask · HTML/CSS/JavaScript

- Web app that checks whether an uploaded image is a black pepper leaf, identifies the nutrient deficiency, and recommends how to mitigate it. Published at SLAAI-ICAI 2024 (see Publications).

Student Dropout Prediction System

TensorFlow · Python · Flask · HTML/CSS/JavaScript

- Predicts whether a newly registered student will graduate, drop out, or stay enrolled, based on registration data entered through a web interface.

AI-Powered Health Symptom Checker

TensorFlow · Python · Flask · HTML/CSS/JavaScript

- Splits symptoms entered by a patient into keywords, predicts the likely disease with a pre-trained ML model, then scrapes the web to present details and medication information.

Other projects: IoT real-time power monitoring system (Spring Boot, Angular, PostgreSQL, C++); wood classification and defect detection (React, Flask, YOLO, PyTorch); recipe management website (React, Spring Boot, OAuth2, JWT, MySQL); automated public space management system for public transport (Pera Innovations, University of Peradeniya); matrix operations processor (VHDL, FPGA).

TECHNICAL SKILLS

AI & LLMs	Claude, AWS Bedrock, Voyage AI, RAG, multi-agent systems, Ollama, Qdrant, ChromaDB
Document AI & Vision	YOLO, TensorFlow, PyTorch, OCR and handwriting extraction
Languages	Python, TypeScript/JavaScript, Java, C++, C, HTML, CSS
Backend & Data	Flask, Django, Spring Boot, PostgreSQL, MySQL, SQLite, AWS SQS/SNS
Frontend	React, Next.js, Angular
Embedded	C++ micro-controllers, VHDL, FPGA, simulation tools

EDUCATION

BSc (Hons) in Engineering, specialised in Computer Engineering – Second Class Upper Division 2018 – 2024
The Open University of Sri Lanka

GCE Advanced Level – Mathematics stream (3 passes) 2017
St. Sylvester's College, Kandy

Following AWS training and certification courses.

PUBLICATION

A. M. S. K. Abeysinghe and G. Anthonys, *"Nutrient Deficiency Detection System for Black Pepper Leaf Using Machine Learning,"* 2024 8th SLAAI International Conference on Artificial Intelligence (SLAAI-ICAI). DOI: 10.1109/SLAAI-ICAI63667.2024.10844967.

AWARDS

Dean's List of Distinguished Students, 2020/2021

LANGUAGES

English, Sinhala

VOLUNTEERING

Students for the Exploration and Development of Space (SEDS) Sri Lanka – Director, IT division

Zero Plastic Planet – Member

REFERENCES

Eng. Dr. G. Anthonys
Senior Lecturer, Department of Electrical and Computer Engineering, The Open University of Sri Lanka, Nawala, Nugegoda
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